

PreACT 8/9 Printable Practice Exam 22

Math • Exam 06

35 questions • 40 minutes

PART A - QUESTIONS

M6-001

What is the value of $-5 + 3^2$?

- A) -14
- B) -4
- C) 4
- D) 8
- E) 14

M6-002

What value of x satisfies $4 - 5x = 24$?

- A) -5
- B) -4
- C) 4
- D) 5
- E) 20

M6-003

Two lines intersect. One of the angles formed measures 126° . What is the measure of an angle adjacent to it?

- A) 36°
- B) 44°
- C) 54°
- D) 126°
- E) 234°

M6-004

A tank contains 72 liters of water and drains at a constant rate, as shown in the table.

t (minutes)	0	3	6	9
V (liters)	72	60	48	36

Which equation gives the volume V after t minutes?

- A) $V = 72 - 4t$
- B) $V = 72 - 12t$
- C) $V = 72 + 4t$
- D) $V = 24 - 4t$
- E) $V = 4t$

M6-005

What is the mode of the data set 6, 8, 8, 9, 11, 11, 11, 14?

- A) 8
- B) 9
- C) 10

- D) 11
- E) 14

M6-006

What is $(7.5 \times 10^8) \div (2.5 \times 10^2)$ written in scientific notation?

- A) 3×10^4
- B) 3×10^6
- C) 3×10^{10}
- D) 5×10^6
- E) 5×10^{10}

M6-007

A \$90 item is discounted by 30%. A 6% sales tax is then applied to the discounted price. What is the final price?

- A) \$59.40
- B) \$63.00
- C) \$64.80
- D) \$66.78
- E) \$68.40

M6-008

A scanner processes 24 pages in 15 minutes at a constant rate. How many pages does it process in 40 minutes?

- A) 49 pages
- B) 60 pages
- C) 64 pages
- D) 72 pages
- E) 96 pages

M6-009

A trapezoid has parallel bases of 8 centimeters and 14 centimeters and a height of 5 centimeters. What is its area?

- A) 44 square centimeters
- B) 50 square centimeters
- C) 55 square centimeters
- D) 70 square centimeters
- E) 110 square centimeters

M6-010

A bag contains 4 yellow counters, 6 purple counters, and 5 white counters. One counter is selected at random. What is the probability that it is purple?

- A) $\frac{1}{5}$
- B) $\frac{4}{15}$
- C) $\frac{1}{3}$
- D) $\frac{2}{5}$
- E) $\frac{3}{5}$

M6-011

What is the value of $\sqrt{225} - 2^3$?

- A) 5

- B) 7
- C) 8
- D) 13
- E) 23

M6-012

The first term of a sequence is 5. Each term after the first is 3 times the preceding term. What is the fifth term?

- A) 45
- B) 135
- C) 243
- D) 405
- E) 1,215

M6-013

A delivery van can carry at most 2,400 kilograms. The driver and equipment have a combined mass of 1,800 kilograms. Each box has a mass of 45 kilograms. What is the greatest number of boxes the van can carry?

- A) 12
- B) 13
- C) 14
- D) 40
- E) 53

M6-014

Point W has coordinates $(-4, 1)$. It is rotated 90° clockwise about the origin. What are the coordinates of the image of W?

- A) $(-1, -4)$
- B) $(-1, 4)$
- C) $(1, -4)$
- D) $(1, 4)$
- E) $(4, 1)$

M6-015

The table shows whether 100 students own a pet.

	Owens a pet	Does not own a pet	Total
Grade 8	27	23	50
Grade 9	33	17	50
Total	60	40	100

One student is selected at random from the students who own a pet. What is the probability that the selected student is in Grade 9?

- A) $17/50$
- B) $9/20$
- C) $11/20$
- D) $3/5$
- E) $33/50$

M6-016

An animal shelter uses 2.4 kilograms of food to feed 16 dogs. At the same rate, how many kilograms of food are needed to feed 30 dogs?

- A) 3.6 kilograms

- B) 4.0 kilograms
- C) 4.5 kilograms
- D) 5.0 kilograms
- E) 7.2 kilograms

M6-017

A café sells 32 items consisting only of sandwiches and drinks. Sandwiches cost \$7, drinks cost \$2, and the total revenue is \$134. How many sandwiches are sold?

- A) 10
- B) 12
- C) 14
- D) 18
- E) 22

M6-018

A culture begins with 50 cells. The number of cells quadruples every 3 hours. How many cells are present after 6 hours?

- A) 200
- B) 400
- C) 600
- D) 800
- E) 3,200

M6-019

A sphere has a radius of 5 centimeters. What is its surface area?

- A) 20π square centimeters
- B) 25π square centimeters
- C) 50π square centimeters
- D) 100π square centimeters
- E) 125π square centimeters

M6-020

Two data sets are shown.

Data Set A: 6, 7, 8, 9, 10 Data Set B: 4, 7, 8, 9, 12

Which statement correctly compares their means and ranges?

- A) The data sets have the same mean and the same range.
- B) The data sets have the same mean, but Data Set A has the greater range.
- C) The data sets have the same mean, but Data Set B has the greater range.
- D) Data Set A has the greater mean and the greater range.
- E) Data Set B has the greater mean and the greater range.

M6-021

What is the value of $6^5 \div (6^2 \times 6)$?

- A) 6
- B) 12
- C) 18
- D) 36
- E) 216

M6-022

The equation $x^2 - 10x + 21 = 0$ has two solutions. What is the greater solution?

- A) 3
- B) 5
- C) 7
- D) 10
- E) 21

M6-023

A printing company charges a setup fee of \$25 plus \$4.50 for each poster. If a customer can spend exactly \$61, how many posters can be ordered?

- A) 6
- B) 7
- C) 8
- D) 9
- E) 13

M6-024

A 12-centimeter-tall model represents an 18-meter-tall building. At the same scale, another model building is 20 centimeters tall. What is the actual height of the second building?

- A) 24 meters
- B) 27 meters
- C) 30 meters
- D) 36 meters
- E) 120 meters

M6-025

Machine A produces items at a constant rate, as shown in the table. Machine B's production after x minutes is $B = 4x - 6$.

x (minutes)	0	4	8
Machine A items	5	17	29

After 10 minutes, how does Machine A's production compare with Machine B's production?

- A) Machine A produces 1 fewer item.
- B) Machine A produces 1 more item.
- C) Machine A produces 5 fewer items.
- D) Machine A produces 5 more items.
- E) The machines produce the same number of items.

M6-026

A rectangular floor is 18 feet long and 14 feet wide. A contractor buys 8% more tile than the floor's area to allow for waste. Each tile covers 1.5 square feet, and only whole tiles can be purchased. What is the least number of tiles needed?

- A) 168
- B) 180
- C) 181
- D) 182
- E) 272

M6-027

What value of x satisfies $7(x - 2) = 3x + 18$?

- A) 4
- B) 6
- C) 8
- D) 10
- E) 14

M6-028

Which of the following numbers is rational?

- A) $\sqrt{13}$
- B) $\sqrt{50}$
- C) $\sqrt[3]{5}$
- D) $\sqrt[3]{8}$
- E) π

M6-029

What is the midpoint of the line segment with endpoints $(-6, 2)$ and $(4, -8)$?

- A) $(-5, -3)$
- B) $(-1, -5)$
- C) $(-1, -3)$
- D) $(1, -3)$
- E) $(5, 3)$

M6-030

A function is defined by $f(x) = 2x + 9$. If $f(k) = -5$, what is the value of k ?

- A) -9
- B) -7
- C) -2
- D) 2
- E) 7

M6-031

Two fair coins are tossed. What is the probability of getting at least one head?

- A) $1/4$
- B) $1/3$
- C) $1/2$
- D) $2/3$
- E) $3/4$

M6-032

A rectangular pool is 5 meters long and 3 meters wide. It contains water to a uniform depth of 0.8 meter. A pump removes water at 0.4 cubic meter per minute. How many minutes will it take to remove all the water?

- A) 12 minutes
- B) 20 minutes
- C) 24 minutes
- D) 30 minutes
- E) 40 minutes

M6-033

A jar contains 45 coins consisting only of quarters and dimes. The coins have a total value of \$7.20. How many dimes are in the jar?

- A) 15
- B) 18
- C) 24
- D) 27
- E) 30

M6-034

A rectangular prism has a volume of 360 cubic centimeters. Its length is 12 centimeters, and its width is 5 centimeters. What is its total surface area?

- A) 216 square centimeters
- B) 252 square centimeters
- C) 300 square centimeters
- D) 324 square centimeters
- E) 360 square centimeters

M6-035

The mean of 8 test scores is 74. One score of 62 is replaced by a score of 86. What is the new mean?

- A) 75
- B) 76
- C) 77
- D) 78
- E) 80

PART B - ANSWER KEY + EXPLANATIONS

M6-001 - Correct: C) 4

Explanation: $3^2 = 9$, so $-5 + 3^2 = -5 + 9 = 4$.

M6-002 - Correct: B) -4

Explanation: Subtract 4 from both sides to get $-5x = 20$, and then divide by -5 to get $x = -4$.

M6-003 - Correct: C) 54°

Explanation: Adjacent angles formed by intersecting lines are supplementary, so the adjacent angle measures $180^\circ - 126^\circ = 54^\circ$.

M6-004 - Correct: A) $V = 72 - 4t$

Explanation: The tank starts with 72 liters and loses 12 liters every 3 minutes, a rate of -4 liters per minute. Therefore, $V = 72 - 4t$.

M6-005 - Correct: D) 11

Explanation: The value 11 appears 3 times, more often than any other value, so the mode is 11.

M6-006 - Correct: B) 3×10^6

Explanation: Divide the coefficients and subtract the exponents: $7.5 \div 2.5 = 3$ and $10^8 \div 10^2 = 10^6$. Therefore, the quotient is 3×10^6 .

M6-007 - Correct: D) \$66.78

Explanation: The discounted price is $\$90 \times 0.70 = \63 . Applying 6% tax gives $\$63 \times 1.06 = \66.78 .

M6-008 - Correct: C) 64 pages

Explanation: The scanner's rate is $24 \div 15 = 1.6$ pages per minute. In 40 minutes it processes $1.6 \times 40 = 64$ pages.

M6-009 - Correct: C) 55 square centimeters

Explanation: The area of a trapezoid is $(1/2)(\text{sum of bases})(\text{height})$, so the area is $(1/2)(8 + 14)(5) = 55$ square centimeters.

M6-010 - Correct: D) $2/5$

Explanation: There are $4 + 6 + 5 = 15$ counters, and 6 are purple. The probability is $6/15 = 2/5$.

M6-011 - Correct: B) 7

Explanation: $\sqrt{225} = 15$ and $2^3 = 8$, so $\sqrt{225} - 2^3 = 15 - 8 = 7$.

M6-012 - Correct: D) 405

Explanation: Multiplying by 3 repeatedly gives 5, 15, 45, 135, 405. Therefore, the fifth term is 405.

M6-013 - Correct: B) 13

Explanation: The remaining carrying capacity is $2,400 - 1,800 = 600$ kilograms. Since $600 \div 45 = 13$ remainder 15, the van can carry at most 13 boxes.

M6-014 - Correct: D) (1, 4)

Explanation: A 90° clockwise rotation maps (x, y) to $(y, -x)$. Therefore, $(-4, 1)$ maps to $(1, 4)$.

M6-015 - Correct: C) $11/20$

Explanation: Of the 60 students who own a pet, 33 are in Grade 9. The probability is $33/60 = 11/20$.

M6-016 - Correct: C) 4.5 kilograms

Explanation: The shelter uses $2.4 \div 16 = 0.15$ kilogram per dog. For 30 dogs, it needs $30 \times 0.15 = 4.5$ kilograms.

M6-017 - Correct: C) 14

Explanation: Let s be sandwiches and d be drinks. Then $s + d = 32$ and $7s + 2d = 134$. Substituting $d = 32 - s$ gives $5s + 64 = 134$, so $s = 14$.

M6-018 - Correct: D) 800

Explanation: Six hours contains 2 three-hour periods. Quadrupling twice gives $50 \times 4^2 = 50 \times 16 = 800$ cells.

M6-019 - Correct: D) 100π square centimeters

Explanation: A sphere's surface area is $4\pi r^2$. With $r = 5$, the area is $4 \times \pi \times 5^2 = 100\pi$ square centimeters.

M6-020 - Correct: C) The data sets have the same mean, but Data Set B has the greater range.

Explanation: Each data set has sum 40 and mean 8. Data Set A has range $10 - 6 = 4$, while Data Set B has range $12 - 4 = 8$, so Data Set B has the greater range.

M6-021 - Correct: D) 36

Explanation: Since $6^2 \times 6 = 6^3$, the expression is $6^5 \div 6^3 = 6^2 = 36$.

M6-022 - Correct: C) 7

Explanation: Factoring gives $x^2 - 10x + 21 = (x - 3)(x - 7)$, so the solutions are 3 and 7. The greater solution is 7.

M6-023 - Correct: C) 8

Explanation: Subtract the \$25 setup fee from \$61 to get \$36. At \$4.50 per poster, the customer can order $\$36 \div \$4.50 = 8$ posters.

M6-024 - Correct: C) 30 meters

Explanation: The scale is 18 meters for 12 centimeters, or 1.5 meters per centimeter. A 20-centimeter model represents $20 \times 1.5 = 30$ meters.

M6-025 - Correct: B) Machine A produces 1 more item.

Explanation: Machine A increases by 12 items every 4 minutes, or 3 items per minute, and starts at 5. After 10 minutes it produces $5 + 3(10) = 35$ items. Machine B produces $4(10) - 6 = 34$ items, so Machine A produces 1 more.

M6-026 - Correct: D) 182

Explanation: The floor area is $18 \times 14 = 252$ square feet. Including 8% extra requires $252 \times 1.08 = 272.16$ square feet. Since $272.16 \div 1.5 = 181.44$, the contractor must buy 182 whole tiles.

M6-027 - Correct: C) 8

Explanation: Expanding gives $7x - 14 = 3x + 18$. Then $4x = 32$, so $x = 8$.

M6-028 - Correct: D) $\sqrt[3]{8}$

Explanation: $\sqrt[3]{8} = 2$, which is an integer and therefore rational. The other roots are not roots of perfect squares or cubes, and π is irrational.

M6-029 - Correct: C) $(-1, -3)$

Explanation: Average the x-coordinates and y-coordinates: $((-6 + 4)/2, (2 + (-8))/2) = (-1, -3)$.

M6-030 - Correct: B) -7

Explanation: Set $2k + 9 = -5$. Then $2k = -14$, so $k = -7$.

M6-031 - Correct: E) $3/4$

Explanation: The 4 equally likely outcomes are HH, HT, TH, and TT. Three outcomes contain at least one head, so the probability is $3/4$.

M6-032 - Correct: D) 30 minutes

Explanation: The pool contains $5 \times 3 \times 0.8 = 12$ cubic meters of water. At 0.4 cubic meter per minute, draining takes $12 \div 0.4 = 30$ minutes.

M6-033 - Correct: D) 27

Explanation: If all 45 coins were dimes, their value would be \$4.50. Each quarter replacing a dime adds \$0.15. The additional $\$7.20 - \$4.50 = \$2.70$ requires $\$2.70 \div \$0.15 = 18$ quarters, leaving $45 - 18 = 27$ dimes.

M6-034 - Correct: D) 324 square centimeters

Explanation: The height is $360 \div (12 \times 5) = 6$ centimeters. The surface area is $2(12 \times 5 + 12 \times 6 + 5 \times 6) = 2(60 + 72 + 30) = 324$ square centimeters.

M6-035 - Correct: C) 77

Explanation: The original total is $8 \times 74 = 592$. Replacing 62 with 86 increases the total by 24, giving 616. The new mean is $616 \div 8 = 77$.